

STAR CLUSTERS

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ALTHOUGH THE STARS in our night sky appear to live out their lives in isolation, many millions of stars reside in groups called open and globular clusters. Open clusters are young and often the site of new star creation, whereas globulars are ancient, dense cities of stars, some of which contain as many stars as a small galaxy.

OPEN STAR CLUSTERS

Open clusters are made up of “sibling” stars of similar age formed from the same nebulous clouds of interstellar gas and dust. This often results in stars within an open cluster having the same chemical composition. However, an open cluster’s stars can exhibit a wide range of masses due to variations within the original nebula and other influences during their formation. Open clusters reside within the galactic plane and often remain associated with the nebulous clouds from which they were produced. Open clusters do not hold onto their stars for long—as they orbit the center of the galaxy, they lose their members over a period of hundreds of millions of years. More than 2,000 open clusters have been discovered within the Milky Way, representing perhaps only 1 percent of the total population.



YOUNG OPEN CLUSTER
Spanning an area in the sky larger than a full moon, M39 is a large but sparsely populated open cluster. It contains about 30 loosely bound stars, each around 300 million years old and therefore much younger than the Sun.



LARGEST CLUSTER
The largest globular cluster in the Milky Way, Omega (ω) Centauri probably contains more than 10 million stars. This makes it larger than some small galaxies.